PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints.(Note- Create Table, insert data as per your Choice)

```
mysql> create table Employee(empId int Not Null, Name varchar(20), emailId varchar(30) unique,
    -> constraint pk_Employee primary key(empId),
    -> constraint un_Employee  unique(emailId));
Query OK, 0 rows affected, 1 warning (0.04 sec)

mysql> create table Department(DeptNo int Not Null, Name varchar(20), emailId varchar(30) unique,
    -> constraint pk_Department primary key(DeptNo),
    -> constraint un_Department unique(emailId));
Query OK, 0 rows affected, 1 warning (0.05 sec)
```

2. Write a SQL query to add any new column to an existing table.

```
mysql> alter table Employee
    -> add column salary int;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

```
mysql>  insert into Employee(empId, Name, emailId,salary)values(1,'Rahul','rahul@gmail.com',50000),
(2,'Ram','ram@gmail.com',55000),(3,'Priya','priya@gmail.com',60000),(4,'Neha','neha@gmail.com',70000),(5,'Nakul','nakul@gmail.com',65000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

mysql> select * from Employee;

```
+-------+-------+-----------------+--------+
| empId | Name  | emailId         | salary |
+-------+-------+-----------------+--------+
|     1 | Rahul | rahul@gmail.com |  50000 |
|     2 | Ram   | ram@gmail.com   |  55000 |
|     3 | Priya | priya@gmail.com |  60000 |
|     4 | Neha  | neha@gmail.com  |  70000 |
|     5 | Nakul | nakul@gmail.com |  65000 |
+-------+-------+-----------------+--------+
```

5 rows in set (0.00 sec)

4. Write a SQL query to delete all records from a table.

mysql> select * from employee;

```
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    | DeptId |
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
|     2 | Ram    | ram@gmail.com   |  60500 | IT         | 2021-06-04   | Kharghar    | 10     |
|     3 | Priya  | priya@gmail.com |  60000 | IT         | 2021-07-20   | Kharghar    | 10     |
|     4 | Neha   | neha@gmail.com  |  70000 | Comp       | 2026-09-06   | Pune        | 20     |
|     5 | Nakul  | nakul@gmail.com |  75000 | Comp       | 2026-07-06   | Pune        | 20     |
|     6 | Sai    | sai@gmail.com   |  74800 | AI         | 2026-07-20   | Juhu        | 30     |
|   101 | Rahul  | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu        | 30     |
|   102 | Anjali | anjali@gmail.com |  50000 | Mech       | 2026-06-02   | Ahilyanagar | 40     |
|   103 | Kiran  | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar | 40     |
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
```

8 rows in set (0.00 sec)

mysql> delete from employee where DeptId=10;

Query OK, 2 rows affected (0.01 sec)

mysql> select * from employee;

```
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    | DeptId |
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
|     4 | Neha   | neha@gmail.com  |  70000 | Comp       | 2026-09-06   | Pune        | 20     |
|     5 | Nakul  | nakul@gmail.com |  75000 | Comp       | 2026-07-06   | Pune        | 20     |
|     6 | Sai    | sai@gmail.com   |  74800 | AI         | 2026-07-20   | Juhu        | 30     |
|   101 | Rahul  | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu        | 30     |
|   102 | Anjali | anjali@gmail.com |  50000 | Mech       | 2026-06-02   | Ahilyanagar | 40     |
|   103 | Kiran  | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar | 40     |
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
```

6 rows in set (0.00 sec)

5. Write a SQL query to update salary records from a table where id = 5.

mysql> update Employee set salary=75000 where empId=5;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from Employee;

```
+-------+-------+-----------------+--------+
| empId | Name  | emailId         | salary |
+-------+-------+-----------------+--------+
|     1 | Rahul | rahul@gmail.com |  50000 |
```

| 2 | Ram | ram@gmail.com | 55000 |

| 3 | Priya | priya@gmail.com | 60000 |

| 4 | Neha | neha@gmail.com | 70000 |

| 5 | Nakul | nakul@gmail.com | 75000 |

+-------+-------+-----------------+--------+

5 rows in set (0.00 sec)

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.

mysql> show grants for 'readonly_user'@'localhost';

+---+

| Grants for readonly_user@localhost |

+---+

| GRANT USAGE ON *.* TO `readonly_user`@`localhost` |

| GRANT SELECT ON `aug26`.`employee` TO `readonly_user`@`localhost` |

+---+

2 rows in set (0.00 sec)

mysql> GRANT SELECT ON Employee TO 'readonly_user'@'localhost';

Query OK, 0 rows affected (0.01 sec)

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.

mysql> create user 'temp_user'@'Localhost';

Query OK, 0 rows affected (0.01 sec)

mysql> show grants for 'temp_user'@'localhost';

+---+

| Grants for temp_user@localhost |

+---+

| GRANT USAGE ON *.* TO `temp_user`@`localhost` |

+--+

1 row in set (0.00 sec)

mysql> grant insert on employee to 'temp_user'@'localhost';

Query OK, 0 rows affected (0.01 sec)

mysql> revoke insert on employee from 'temp_user'@'localhost';

Query OK, 0 rows affected (0.01 sec)

8. Write a query to insert a new employee into the Emp table and commit the transaction.

mysql> start transaction;

Query OK, 0 rows affected (0.00 sec)

mysql> insert into Employee(empId, Name, emailId,salary)values(6,'Sai','sai@gmail.com',68000);

Query OK, 1 row affected (0.00 sec)

mysql> commit;

Query OK, 0 rows affected (0.01 sec)

mysql> select * from Employee;

+-------+-------+-----------------+--------+

| empId | Name | emailId | salary |

+-------+-------+-----------------+--------+

| 1 | Rahul | rahul@gmail.com | 50000 |

| 2 | Ram | ram@gmail.com | 55000 |

| 3 | Priya | priya@gmail.com | 60000 |

| 4 | Neha | neha@gmail.com | 70000 |

| 5 | Nakul | nakul@gmail.com | 75000 |

| 6 | Sai | sai@gmail.com | 68000 |

+-------+-------+-----------------+--------+

6 rows in set (0.00 sec)

9. Write a query to update the Salary of an employee with EmpID = 101 and commit the transaction.

mysql> start transaction;

Query OK, 0 rows affected (0.00 sec)

mysql> update Employee set salary=56000,empId=1 where empId=1;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> commit;

Query OK, 0 rows affected (0.01 sec)

mysql> select * from Employee;

+-------+-------+-----------------+--------+

| empId | Name | emailId | salary |

+-------+-------+-----------------+--------+

| 1 | Rahul | rahul@gmail.com | 56000 |

| 2 | Ram | ram@gmail.com | 55000 |

| 3 | Priya | priya@gmail.com | 60000 |

| 4 | Neha | neha@gmail.com | 70000 |

| 5 | Nakul | nakul@gmail.com | 75000 |

| 6 | Sai | sai@gmail.com | 68000 |

```
+-------+-------+-----------------+---------+
```

6 rows in set (0.00 sec)

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint s1;
Query OK, 0 rows affected (0.00 sec)

mysql> update Employee set DeptId=5 where empId=2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Employee;
+-------+-------+-----------------+--------+---------+
| empId | Name  | emailId         | salary | DeptId |
+-------+-------+-----------------+--------+---------+
|     2 | Ram   | ram@gmail.com   |  55000 |      5 |
|     3 | Priya | priya@gmail.com |  60000 |   NULL |
|     4 | Neha  | neha@gmail.com  |  70000 |   NULL |
|     5 | Nakul | nakul@gmail.com |  75000 |   NULL |
|     6 | Sai   | sai@gmail.com   |  68000 |   NULL |
|   101 | Rahul | rahul@gmail.com |  58000 |   NULL |
+-------+-------+-----------------+--------+---------+
6 rows in set (0.00 sec)
```

11. Write a query to rollback to the previously created SAVEPOINT.

```
mysql> rollback to s1;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from Employee;
+-------+-------+-----------------+--------+--------+
| empId | Name  | emailId         | salary | DeptId |
+-------+-------+-----------------+--------+--------+
|     2 | Ram   | ram@gmail.com   |  55000 |   NULL |
|     3 | Priya | priya@gmail.com |  60000 |   NULL |
|     4 | Neha  | neha@gmail.com  |  70000 |   NULL |
|     5 | Nakul | nakul@gmail.com |  75000 |   NULL |
|     6 | Sai   | sai@gmail.com   |  68000 |   NULL |
|   101 | Rahul | rahul@gmail.com |  58000 |   NULL |
+-------+-------+-----------------+--------+--------+
6 rows in set (0.00 sec)
```

12. Write a query to truncate all data from the Emp table.

```
mysql> truncate department;
Query OK, 0 rows affected (0.05 sec)

mysql> select * from department;
Empty set (0.00 sec)
```

13. Write a query to list all employees whose salary is greater than 50,000.

```
mysql> select empId,name,salary from Employee where salary > 5000;
+-------+-------+--------+
| empId | name  | salary |
+-------+-------+--------+
```

| 2 | Ram | 55000 |

| 3 | Priya | 60000 |

| 4 | Neha | 70000 |

| 5 | Nakul | 75000 |

| 6 | Sai | 68000 |

| 101 | Rahul | 58000 |

+-------+-------+--------+

6 rows in set (0.00 sec)

14. Write a query to increase the salary of all employees in DeptID = 10 by 10%.

mysql> update Employee set salary=salary*1.10 where DeptId=2;

Query OK, 0 rows affected (0.00 sec)

Rows matched: 0 Changed: 0 Warnings: 0

mysql> select * from Employee;

+-------+-------+-----------------+--------+--------+

| empId | Name | emailId | salary | DeptId |

+-------+-------+-----------------+--------+--------+

| 2 | Ram | ram@gmail.com | 55000 | NULL |

| 3 | Priya | priya@gmail.com | 60000 | NULL |

| 4 | Neha | neha@gmail.com | 70000 | NULL |

| 5 | Nakul | nakul@gmail.com | 75000 | NULL |

| 6 | Sai | sai@gmail.com | 68000 | NULL |

| 101 | Rahul | rahul@gmail.com | 58000 | NULL |

+-------+-------+-----------------+--------+--------+

6 rows in set (0.00 sec)

15. Write a query to count the total number of employees in each department.

```
mysql> select DeptId, count(empId) from Employee group by DeptId;

+--------+--------------+

| DeptId | count(empId) |

+--------+--------------+

|      5 |            4 |

|     10 |            2 |

+--------+--------------+

2 rows in set (0.00 sec)
```

16. Write a query to list departments that have no employees. & create a view for those salespeople who belong to the city of Mumbai.

```
mysql> select DepartmentName from department

   -> where DeptNo NOT IN(select distinct DeptId from Employee where deptNo is NOT
NULL);

+----------------+

| DepartmentName |

+----------------+

| IT             |

+----------------+

1 row in set (0.01 sec)

mysql> create view mumbai_salespeople as select empId, Name, emailId, salary,
Department, location

   -> from Employee

   -> where location='Kharghar';

Query OK, 0 rows affected (0.01 sec)


mysql> select * from mumbai_salespeople;

+-------+-------+-----------------+--------+------------+----------+
```

| empId | Name | emailId | salary | Department | location |

+-------+-------+-----------------+--------+------------+----------+

| 2 | Ram | ram@gmail.com | 60500 | IT | Kharghar |

| 3 | Priya | priya@gmail.com | 60000 | IT | Kharghar |

+-------+-------+-----------------+--------+------------+----------+

2 rows in set (0.00 sec)

17. Write a query to display the highest salary among all employees.

mysql> select max(salary) from Employee;

+-------------+

| max(salary) |

+-------------+

| 75000 |

+-------------+

1 row in set (0.00 sec)

18. Write a query to display the 2nd highest salary among all employees.

mysql> select * from Employee order by salary desc limit 1 offset 1;

+-------+------+---------------+--------+--------+

| empId | Name | emailId | salary | DeptId |

+-------+------+---------------+--------+--------+

| 6 | Sai | sai@gmail.com | 74800 | 10 |

+-------+------+---------------+--------+--------+

1 row in set (0.00 sec)

19. Write a query to show all employees ordered by JoiningDate in descending order.

mysql> select * from Employee order by Joining_Date desc;

+-------+-------+-----------------+--------+--------+--------------+

| empId | Name | emailId | salary | DeptId | Joining_Date |

```
+-------+-------+-----------------+--------+--------+--------------+
|     3 | Priya | priya@gmail.com |  60000 |      5 | 2026-10-06   |
|     2 | Ram   | ram@gmail.com   |  60500 |      5 | 2026-10-04   |
|   101 | Rahul | rahul@gmail.com |  63800 |     10 | 2026-09-18   |
|     4 | Neha  | neha@gmail.com  |  70000 |      5 | 2026-09-06   |
|     6 | Sai   | sai@gmail.com   |  74800 |     10 | 2026-07-20   |
|     5 | Nakul | nakul@gmail.com |  75000 |      5 | 2026-07-06   |
+-------+-------+-----------------+--------+--------+--------------+
6 rows in set (0.00 sec)
```

20. Write a query to rename the DeptName column to DepartmentName.

```
mysql> alter table department
   -> rename column DeptName to DepartmentName;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0


mysql> select * from department;
+--------+----------------+---------------------+
| DeptNo | DepartmentName | emailId             |
+--------+----------------+---------------------+
|      5 | IT             | it@company.com      |
|     20 | HR             | hr@company.com      |
|     30 | Finance        | finance@company.com |
+--------+----------------+---------------------+
3 rows in set (0.00 sec)
```